Tutorial week 4

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1. Model Selection Criteria

Question

A dataset contains $n = 100$ observations. Using best subset selection, three candidate regression models are obtained:

Model	Number of Predictors (p)	RSS
M_1	3	220
M_2	5	200
M_3	8	180

The full model (with 10 predictors) has $RSS_{full} = 170$. Let

$$\hat{\sigma}^2 = \frac{RSS_{full}}{n - 10 - 1}, \qquad TSS = 500.$$

(a) Compute Mallows' C_p for each model. Which model would you select?

(b) Compute AIC for each model. Which model would you select?

(b) Compute BIC for each model. Which model would you select?

(c) Compute Adjusted R^2 for each model. Which model would you select?

(e) Explain mathematically why BIC tends to prefer smaller models than AIC when n is large.

Answer

Step 1: Estimate Noise Variance

$$\hat{\sigma}^2 = \frac{170}{100 - 10 - 1} = \frac{170}{89} = 1.91.$$

(a) Mallows' C_p

$$C_p = \frac{1}{n}(RSS + 2p\sigma^2)$$

$$M_1 : 2.315 \qquad M_2 : 2.191 \qquad M_3 : 2.106$$

$$\boxed{C_p \text{ selects } M_3}$$

(b) AIC

$$AIC = n \log\left(\frac{RSS}{n}\right) + 2p$$

$$M_1 : 84.85 \qquad M_2 : 79.31 \qquad M_3 : 74.78$$

$$\boxed{\text{AIC selects } M_3}$$

(c) BIC

$$BIC = n \log\left(\frac{RSS}{n}\right) + p \log n$$

$$M_1 : 92.67 \qquad M_2 : 92.34 \qquad M_3 : 95.62$$

$$\boxed{\text{BIC selects } M_2}$$

(d) Adjusted R^2

$$\bar{R}^2 = 1 - \frac{RSS/(n - p - 1)}{TSS/(n - 1)}$$

$$M_1 : 0.546 \qquad M_2 : 0.579 \qquad M_3 : 0.608$$

$$\boxed{\text{Adjusted } R^2 \text{ selects } M_3}$$

(e) Why BIC Prefers Smaller Models

AIC penalty: $2p$
BIC penalty: $p \log n$

Since for $n > 7$,

$$\log n > 2,$$

BIC penalizes model complexity more heavily, and therefore tends to choose simpler (smaller) models.

2. K-Fold Cross-Validation

Question

Suppose a dataset contains $n = 100$ observations and is divided into $K = 5$ equal folds. A regression model is evaluated using 5-fold cross-validation.

The validation Mean Squared Errors (MSE) obtained are:

$$4.2, \quad 5.0, \quad 4.8, \quad 5.5, \quad 4.5.$$

(a) Write the mathematical formula for K-fold cross-validation error.

(b) Compute the 5-fold CV estimate.

(c) How many models are trained?

(d) If $K = 100$ (LOOCV), how many models are trained?

(e) Compare LOOCV and 5-fold CV with respect to bias and variance.

Answer

(a) Formula

$$CV(K) = \sum_{k=1}^{K} \frac{n_k}{n} MSE_k$$

If folds are equal:

$$CV(K) = \frac{1}{K} \sum_{k=1}^{K} MSE_k$$

(b) Compute 5-Fold CV Error

$$CV(5) = \frac{1}{5}(4.2 + 5.0 + 4.8 + 5.5 + 4.5) = \frac{24}{5} = 4.8$$

$$\boxed{CV(5) = 4.8}$$

(c) Number of Models

$$\boxed{\text{5 models}}$$

(d) LOOCV

If $K = n = 100$:

$$\boxed{\text{100 models}}$$

(e) Bias-Variance Tradeoff

LOOCV trains on $n - 1$ observations, which is very close to n, so:

$$\text{Bias(LOOCV)} < \text{Bias(5-fold)}.$$

However, training sets in LOOCV differ by only one observation, making validation errors highly correlated, which increases variance:

$$\text{Var(LOOCV)} > \text{Var(5-fold)}.$$

Gradient Descent: Questions and Solutions

Question 1

Consider the one-dimensional quadratic function:

$$f(\theta) = a\theta^2, \qquad a > 0.$$

1. Derive the gradient descent update rule.
2. Determine the condition on the learning rate α for convergence to the minimum.

Solution

Step 1: Compute the gradient

$$\frac{df}{d\theta} = 2a\theta.$$

Step 2: Gradient Descent Update Rule

$$\theta_{t+1} = \theta_t - \alpha \frac{df}{d\theta} = \theta_t - \alpha(2a\theta_t).$$

$$\theta_{t+1} = (1 - 2a\alpha)\theta_t.$$

Step 3: Convergence Condition
For convergence to zero, we require:

$$|1 - 2a\alpha| < 1.$$

This gives:

$$-1 < 1 - 2a\alpha < 1.$$

Solving,

$$0 < \alpha < \frac{1}{a}.$$

$$\boxed{\text{Gradient descent converges if } 0 < \alpha < \tfrac{1}{a}.}$$

Question 2

Consider the function:

$$J(\beta) = (\beta - 3)^2.$$

Starting from $\beta_0 = 0$ and using learning rate $\alpha = 0.1$:
1. Derive the gradient.
2. Compute β_1 and β_2.

Solution

Step 1: Compute Gradient

$$\frac{dJ}{d\beta} = 2(\beta - 3).$$

Step 2: Gradient Descent Update Rule

$$\beta_{t+1} = \beta_t - \alpha 2(\beta_t - 3).$$

Step 3: First Iteration

$$\beta_1 = 0 - 0.1 \cdot 2(0 - 3) = 0 - 0.1(-6) = 0.6.$$

Step 4: Second Iteration

$$\beta_2 = 0.6 - 0.1 \cdot 2(0.6 - 3) = 0.6 - 0.1(-4.8) = 0.6 + 0.48 = 1.08.$$

$$\boxed{\beta_1 = 0.6, \quad \beta_2 = 1.08.}$$

The iterates move toward the true minimum at $\beta = 3$.

Question 3 (Try at home)

Let

$$f(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2 + \theta_1\theta_2$$

Check whether f is convex. If yes, find the global minima and the global minimum value directly i.e. without gradient descent. Write the gradient descent update rule. Find first 2 iterates, where initial point taken is $(\theta_1^{(0)}, \theta_2^{(0)}) = (1, -1)$ and step size $\alpha = 0.001$.